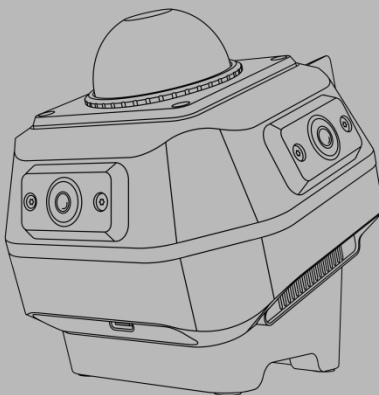


MindPalace

Pocket 2 & Pocket 2pro

USER MANUAL

V0.1.4



Manifold Tech Limited

2026.2

EN

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1. Introduction

Pocket2 is a professional-grade, high-precision 3D data acquisition and intelligent processing device that seamlessly integrates point cloud scanning, panoramic imaging, and detailed photo recording in one compact unit. Powered by the high-performance MindSLAM™ algorithm, the device enables real-time 3D data acquisition and processing, significantly improving on-site operational efficiency. Pocket2 is widely applicable in digital twin construction, facility operation and maintenance, and robotic simulation training, while also serving firefighting, public security, transportation, and forestry sectors for emergency response and investigation. Through deep integration with the MindCloud platform, Pocket2 supports seamless generation of 3D models from 2D imagery, enabling a fully connected digital twin workflow — from data acquisition to computation and modeling.

The Pocket2 Pro version builds upon the standard model with an additional dual top-mounted 12-megapixel panoramic camera module, delivering higher-resolution scene capture and superior 3DGS rendering performance. It meets the demanding standards of professional users who require enhanced visual fidelity, richer spatial details, and optimized data quality for advanced 3D applications.

2. Basic Functions of Equipment

2.1 Physical Structure of Pocket 2 & Pocket 2Pro

Pocket 2



Pocket 2 Pro



- | | | |
|-----------------------------------|---------------------------------|--------------------------------|
| 1. LIDAR | 5. Data Transfer Port | |
| 2. SLAM Camera | 6. Power Button/Indicator Light | 9. Dual 12MP Panoramic Cameras |
| 3. External Camera Shutter Button | 7. Charging Port | |
| 4. Expansion Hot Shoe Port | 8. External Macro Camera | |

*The external macro camera is an optional accessory.

*The dual 12MP cameras are standard for Pocket 2 Pro, Pocket 2 does not include this component.

2.2 Battery Installation and Charging

Press the latch buttons indicated by the red box simultaneously to attach and secure the handle battery to the device via the latch connection.



As shown below, press and hold the power button to start the device (a short press checks the battery level). Wait for the power indicator lights to illuminate sequentially.



2.3 Indicator Lights and Buttons

Battery indicator light status	Meaning
LED 1 is blinking blue.	The battery is charging, with a charge level ranging from 0-25%

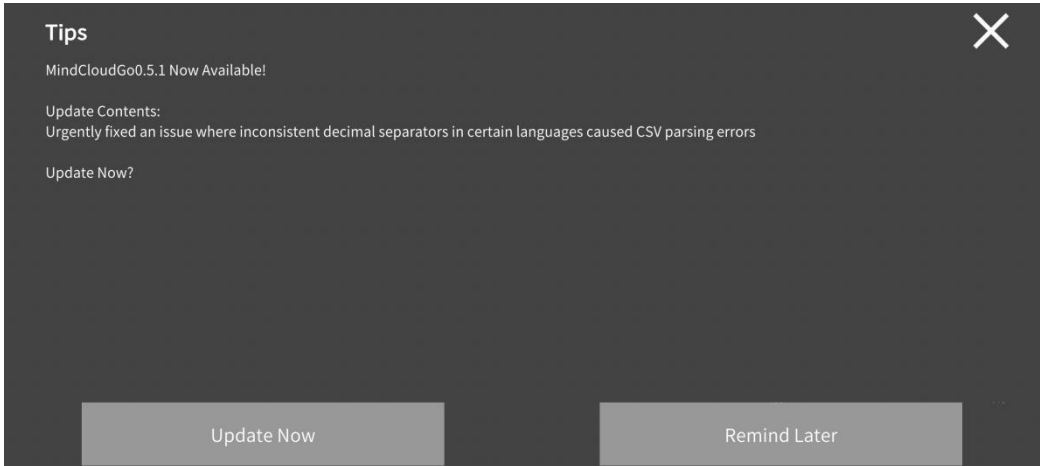
LED 1 is solid blue. LED 2 is blinking blue.	The battery is charging, with a charge level ranging from 25-50%
LEDs 1 and 2 are solid blue. LED 3 is blinking blue.	The battery is charging, with a charge level ranging from 50-75%
LEDs 1, 2, and 3 are solid blue. LED 4 is blinking blue.	The battery is charging, with a charge level ranging from 75-95%
LEDs 1, 2, 3 and 4 are solid blue.	The battery is charging and the battery level is over 95%; The battery is discharging, with a charge level ranging from 100% to 75%
LEDs 1, 2 and 3 are solid blue.	The battery is discharging, with a charge level ranging from 75-50%
LEDs 1 and 2 are solid blue.	The battery is discharging, with a charge level ranging from 50-25%
LED 1 is solid blue.	The battery is discharging, with a charge level ranging from 25-7%
LED 1 is blinking red.	The battery is discharging, with a charge level ranging from 7-5%
LEDs 1, 2, 3 and 4 are blinking blue.	Battery Overtemperature Warning
Button mode	Function
Press and hold for 3 seconds	Switch the device between on and off states
Press once briefly	Check battery level

2.4 Data Copy

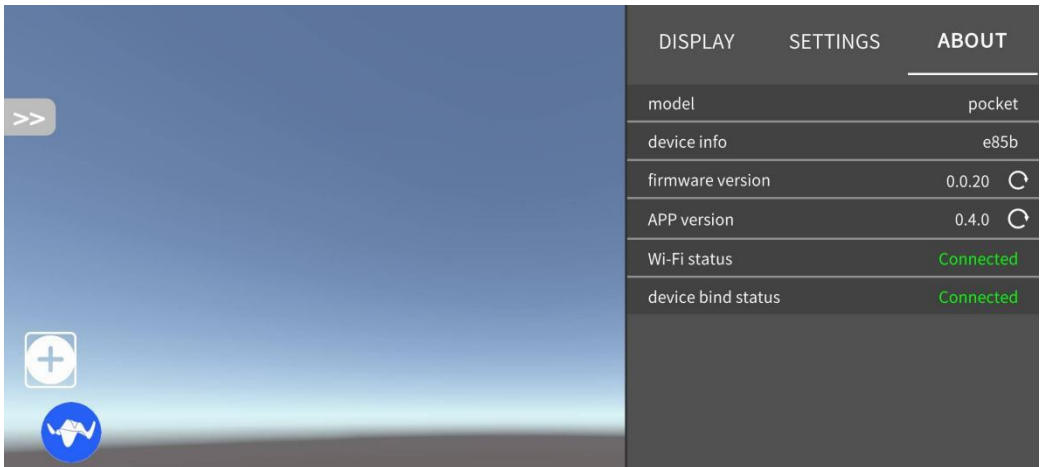
After pressing and holding the device power button to turn it on, use the supplied Type-C data cable to connect the device's DATA port to the computer. Once the device storage is recognized, you can proceed with data copying.

2.5 APP and Firmware Upgrade

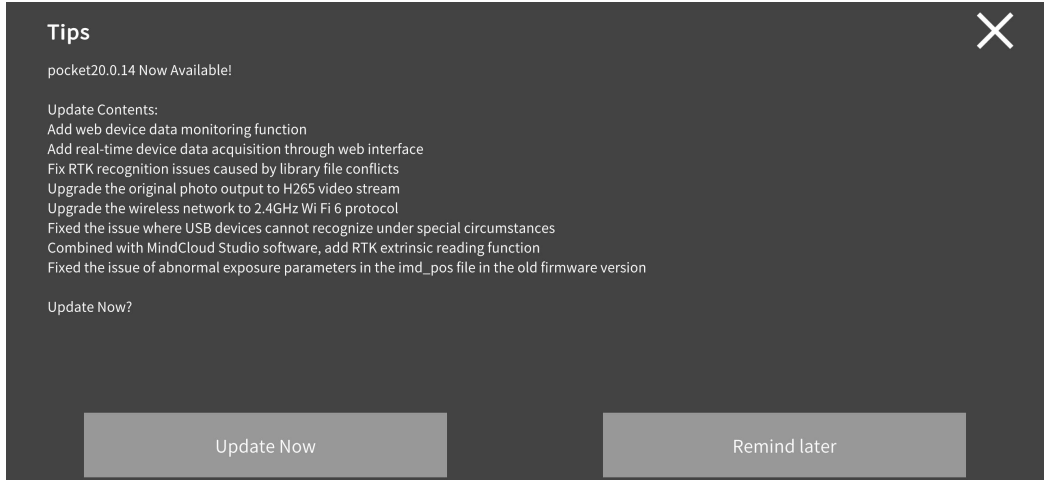
After the user opens the APP, the APP will automatically check whether the current version is the latest version. If it is not the latest version, a pop-up prompt will be displayed.



Users can also manually check the APP version by entering the "Start" - "Settings" - "About" page and clicking the refresh icon on the right side of the APP version column to manually check the version status.



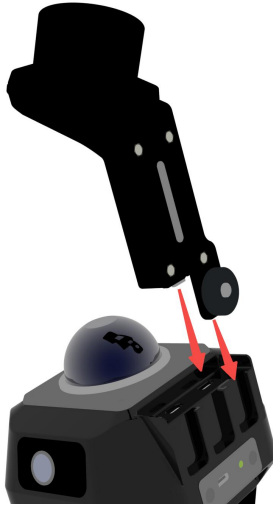
After connecting to the device via the APP, the APP automatically checks whether the device firmware is up to date. If not, a pop-up notification will appear, allowing the user to choose whether to perform an online firmware update.



After the online firmware update is completed, the device and the APP need to be restarted before use.

2.6 RTK Installation and Configuration

Insert the RTK connector into the device's reserved Port C. The RTK extension bracket will simultaneously slide into the hot shoe slot. Once you confirm that both connectors are fully seated, tighten the securing knob to complete the RTK installation on the Pocket 2.

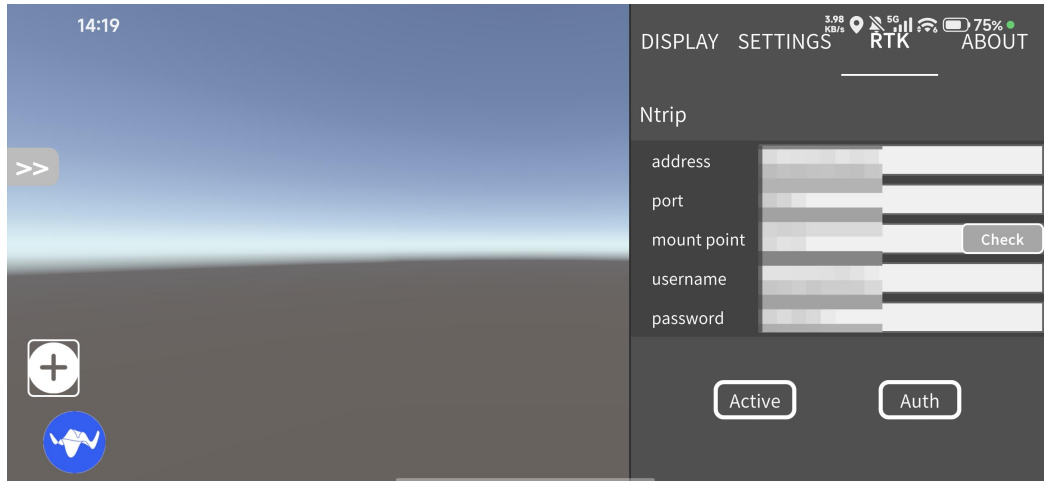


The RTK for the Pocket 2 Pro is mounted and secured to the device using an RTK bracket, as shown in the figure below.



Enter the configuration information obtained from your CORS service

provider into the RTK settings, then click "Authenticate." Once the "Ntrip authentication successful" pop-up appears, click "Enable." After the "Ntrip service started" message is displayed, the RTK configuration is complete.



After starting the scan, pay attention to the color change of the RTK icon. A green icon indicates centimeter-level accuracy and confirms that an RTK fixed solution has been obtained.



2.7 3DGS Configuration and Usage

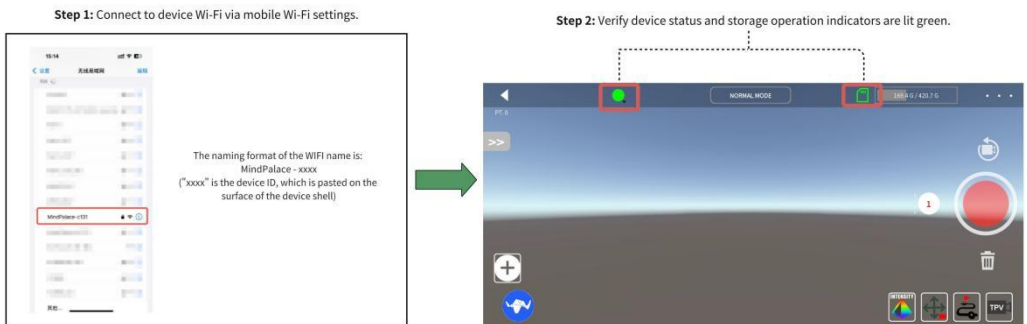
Currently, only the Pocket2 Pro supports the 3DGS. For detailed configuration and usage, please refer to the [MindCloud Renderer 使用说明](#).

3. Data Acquisition Process

3.1 MindCloudGo Introduction

MindCloudGo is the companion mobile application for the Mindpalace Pocket2. By connecting to the device via WiFi, it allows real-time point cloud viewing and supports quick switching between true color, intensity, and elevation modes. Users can monitor scanning and mapping results as well as walking trajectories in real time, adjust scanning angles and positions flexibly to prevent missed areas, and efficiently complete 3D scanning tasks. In addition, MindCloudGo supports local and device project management and viewing, as well as uploading and exporting point cloud data, enabling intelligent spatial digital management in the cloud.

3.2 Device Connection



Power on the device, connect your mobile phone to the corresponding device WiFi (WiFi name format: MindPalace-xxxx, password: 1234567890),

and launch the MindCloudGo app to start data acquisition. If the WiFi cannot be connected or the app shows “Not Connected,” please check and try the following:

- Disable mobile data on your phone’s SIM card;
- When connecting to Wi-Fi, select options like "Trust this Wi-Fi" or "Keep Connected" to maintain a stable connection;
- Disable network proxy or traffic-intercepting software like VPN(if enabled);
- Turn off advanced WLAN settings such as "Dual Wi-Fi Acceleration", "Smart Wi-Fi Selection", "Data Network Boost" or "Auto-Switch Data Network" (if enabled).

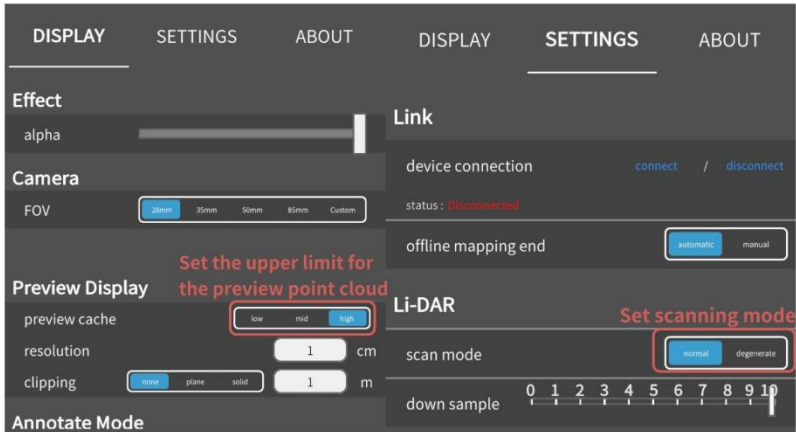
3.3 Scanning Operation

On the main interface, click "Start" to enter the scanning page.



Enter the settings page and switch the scanning mode according to the actual situation. If the phone performance is poor, please adjust the upper limit of the preview point cloud to a lower value.

⚠ Except for the radar scanning mode, other settings will not affect the original data.



Place the device on a relatively level surface and click "Start Recording".

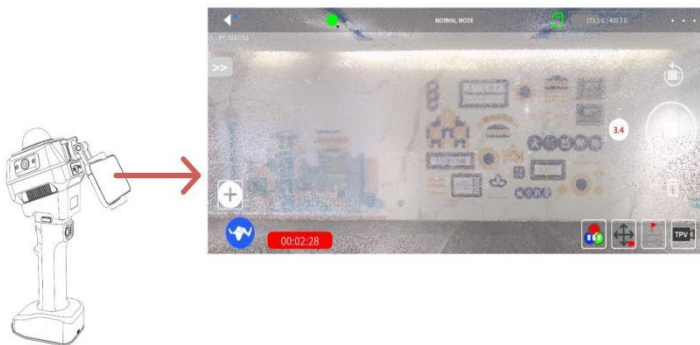


Once the initialization prompt disappears, you can pick up and move the device for scanning.

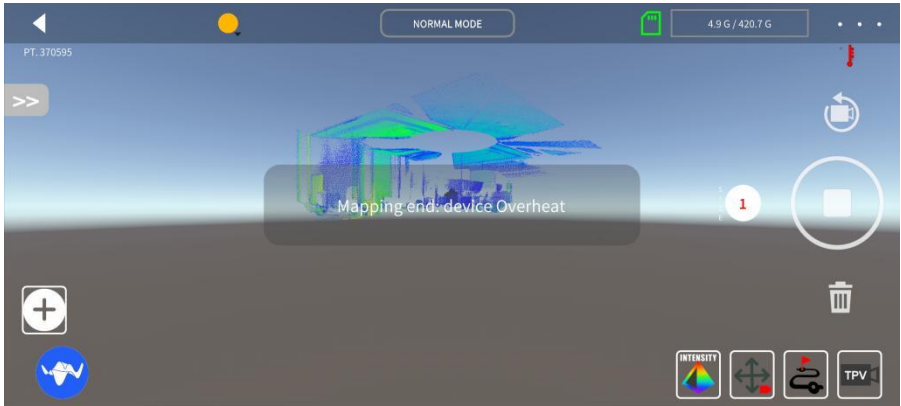
⚠ Do not move the device before the initialization prompt window disappears.



During scanning, hold the device vertically to the ground, keeping it in front of your body. Monitor point cloud completeness via the APP preview interface and rescan any missing areas promptly.

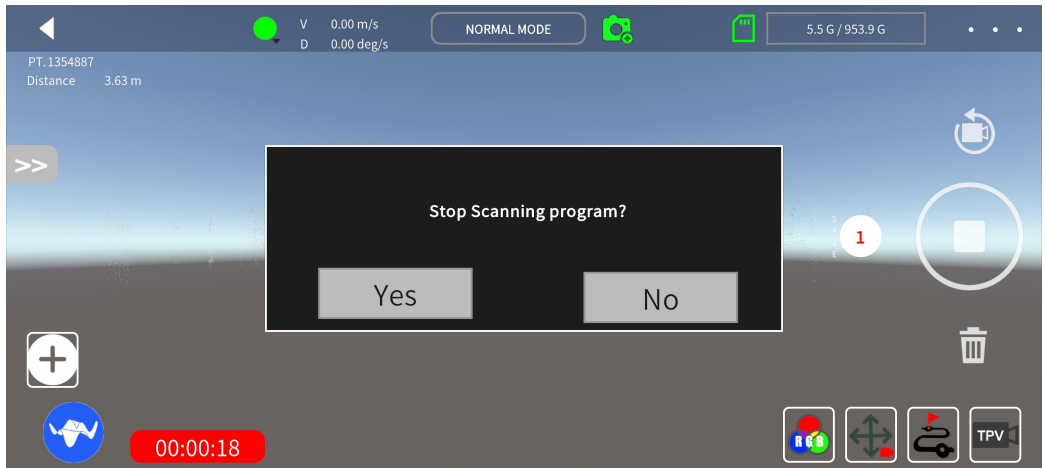


In rare cases during prolonged outdoor scanning under high temperatures, the app may display a warning as shown in the example below. In this situation, turn off the device and let it sit until the temperature decreases, then resume scanning.



After scanning, click "Stop Recording" to end the scan.

⚠ The device is prohibited from being powered off or disconnected from the power supply until the file save prompt window disappears.



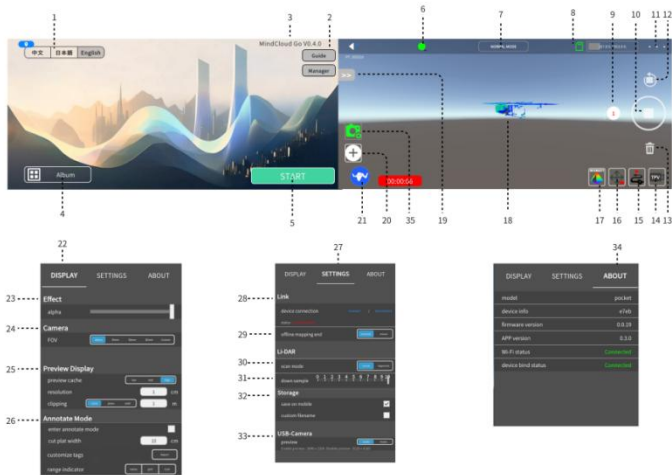


3.4 Precautions for Use

- After connecting your mobile phone to the device WiFi, a network unavailable warning may appear. Do not switch WiFi networks.
- Avoid sudden shaking or rotation of the device during data acquisition.
- During prolonged use, do not touch or cover the device's heat dissipation areas.

4. Detailed App Functions

4.1 APP Interface Diagram



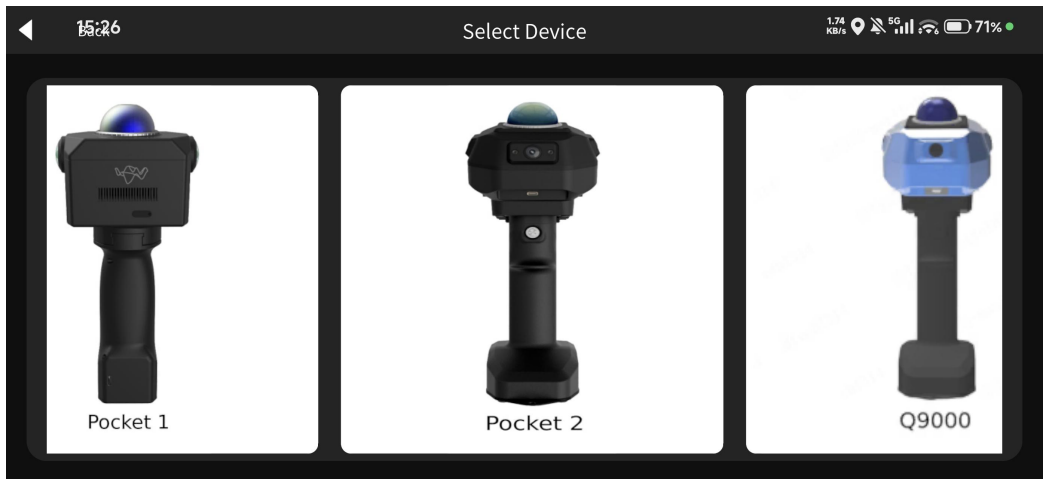
1. Language switch
2. User manual/log
3. APP version
4. Historical project set
5. Enter scanning control interface
6. Device status
7. Radar scanning mode
8. SSD status and storage
9. Preview point cloud size
10. Start/stop recording
11. Settings menu
12. Restore initial view
13. Clear the preview point cloud
14. Switch camera view
15. Show/hide trajectory
16. Follow mode / Wander mode
17. Switch point cloud coloring mode
18. Point cloud preview window
19. Viewpoint movement virtual joystick
20. Control point acquisition
21. Point cloud preview window full screen
22. Display Settings
23. Point cloud preview transparency
24. Point cloud preview window viewing range
25. Preview Settings
 - Preview point cloud quantity limit: This refers to the maximum number of point clouds that can be displayed in the preview window. It can be adjusted according to the performance of the mobile phone.
 - Preview point cloud point spacing: The number represents the point cloud density reduction distance. The larger the number, the greater the point spacing. The range is 1 to 10.
 - Preview point cloud by distance clipping: It can be clipped according to plane/solid. The number represents the distance in front of the device as the origin, in meters.
26. Annotation Mode
 - Enable/Disable Annotation Mode
 - Section Thickness: The number represents the thickness of the point cloud plane slice
 - Custom Labels: Can import a label text set, which can be used to quickly select the desired label name during annotation
 - Range Indicators: Three methods for representing range - optional radius, grid, and ruler.
27. Collection settings
28. Device connection status
29. Offline mapping mode
 - Automatic: The scan will automatically end 15 seconds after the APP is disconnected from the device.
 - Manual: After starting the scan, you can disconnect the phone from the device. You must reconnect and then click the "End Recording" button to end the scan.
30. Scanning mode
 - Degraded mode: Suitable for scenarios with highly consistent environmental characteristics such as tunnels, long corridors, and underground passages.
 - normal mode: Suitable for the vast majority of environments except for the degraded scenarios.
31. Downsampling ratio: Adjust the proportion of the number of receiving points in the mobile device. This setting is made before scanning based on the performance of the mobile device.
32. Storage
 - Store to mobile phone: Once enabled, it will store the previewed point cloud locally on the mobile device and supports offline viewing in the history project list.
 - Customize file name: Once enabled, it supports custom naming of the file before starting the scan.
33. Macro Camera Preview Video Switch
 - On: Supports camera preview, but photos are captured in 4K resolution.
 - Off: Disables video preview; photos are captured in 8K resolution and support photo preview.
34. Equipment Information

*The macro camera preview toggle option will only appear after the macro camera hardware is installed.

4.2 APP Home Page Introduction

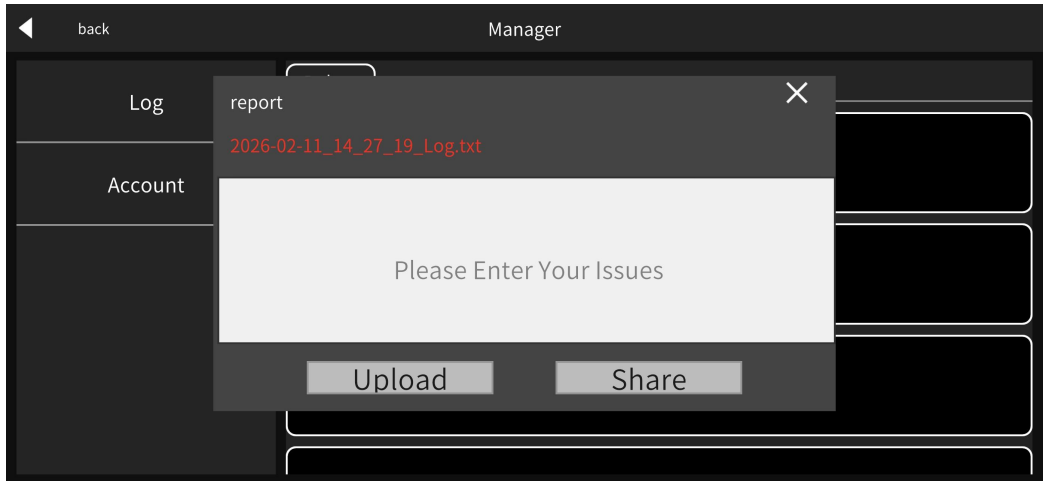


Click the 'Guide' option at the top right of the homepage to enter the device selection page. When connected to the internet, users can choose to cache and open the latest version of the user manual for the corresponding product. Without a data connection, the APP's built-in user manual will be opened.

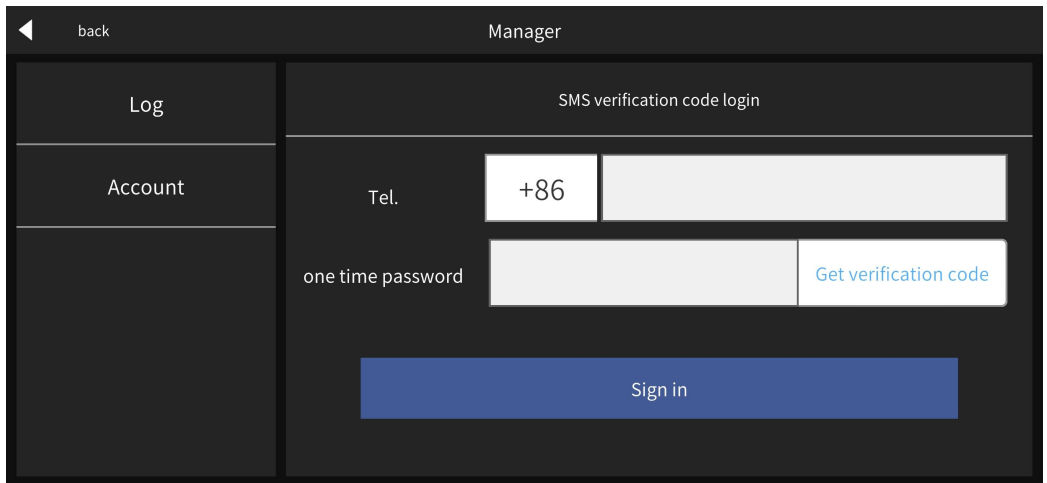


Tap the "Manager" option in the top-right corner of the home screen, then tap "Log" to enter the log management page. When connected to the internet, users can select the relevant project log file, describe the issue encountered, and

tap "Upload" to send both the description and log file to the server. Tapping "Share" allows the log file to be shared to other platforms.

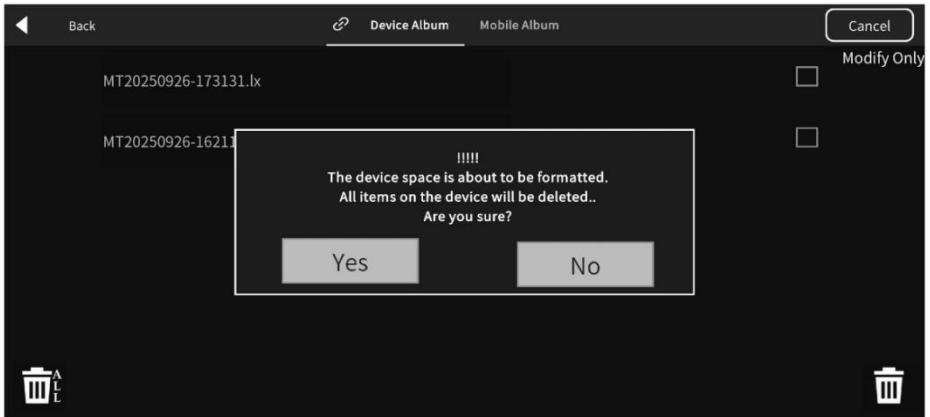


Tap the "Manager" option in the top-right corner of the home screen, then tap "Account." You can obtain a login verification code via your mobile number, and after logging in, you' ll have access to all APP features.



Click the 'Album' option at the bottom left of the homepage to enter the project management page. Select the 'Delete' option at the bottom right to delete

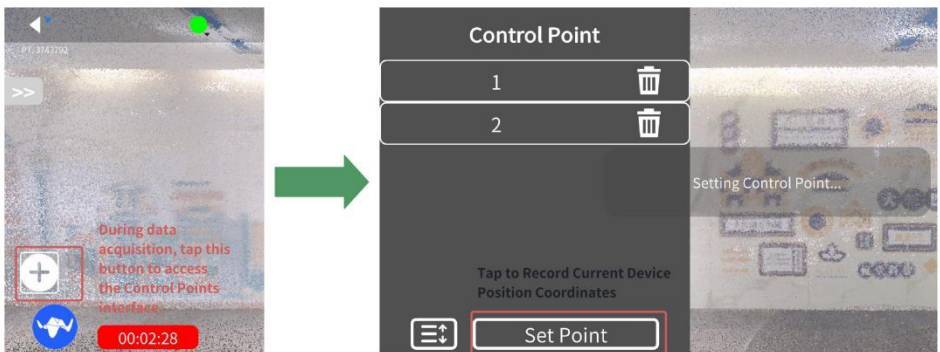
the selected project files stored on the device. Choose the 'Delete All' option at the bottom left to delete all project files stored on the device with one click.



4.3 Control Point Function

During scanning, you can collect coordinates of known ground points for post-processing point cloud coordinate transformation.

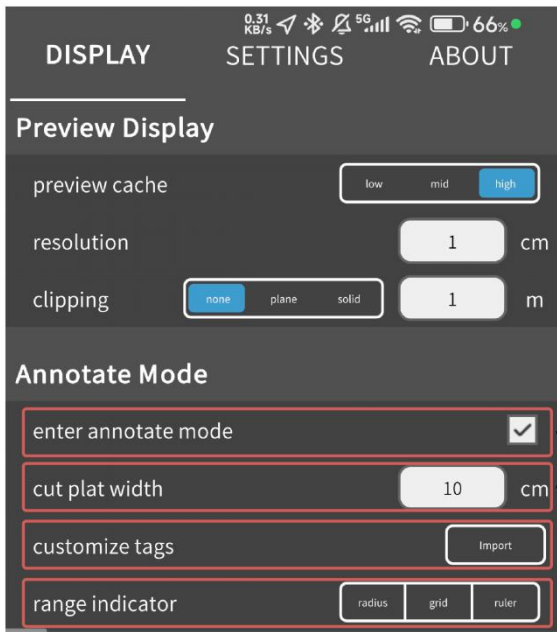
- ⚠ Before collecting the control points, please align the observation hole at the bottom of the equipment with the center of the ground target.



4.4 Annotation Function

This function is useful for forest scenarios or when recording spatial coordinates of specific objects. Before scanning, enable annotation mode via "Display" "Annotate Mode".

⚠ After enabling this function, the point cloud preview will default to the topdown view, and the point cloud will be sectioned according to the preset thickness. After the scan is completed, a.csv text file containing the recorded annotation information will be generated in the result folder.



Tick this option to activate Annotation Mode

Set Profile Thickness

Import Label Library File

Three methods to represent distance



4.5 Offline Mapping

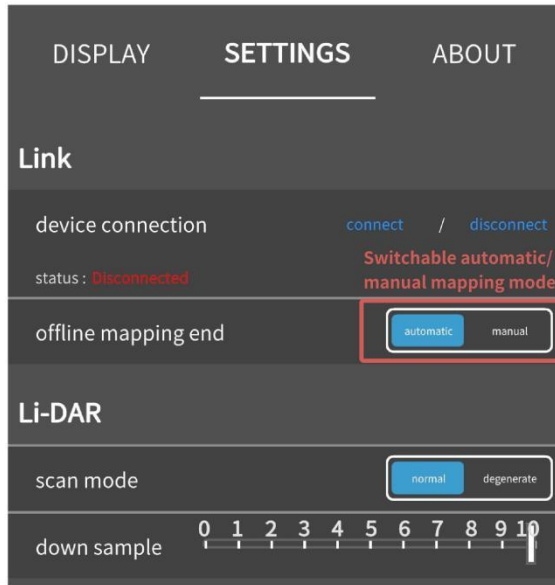
This function is for scenarios requiring scanning without a phone connection.

Automatic Mode: After scanning starts, disconnecting the phone for 15 seconds ends the scan automatically.

Manual Mode: Disconnecting the phone does not end the scan; you must reconnect and end it via the APP.

⚠ Please switch this mode before the scanning begins.

In manual mode, the scanning process will not be terminated except by clicking the "End Scan" button in the APP. In any other situation, the scanning will continue. If the device runs out of battery and is shut down before clicking the "End" button in the APP, the current scanning data will not be saved.



4.6 Macro Camera Photography

Install the macro camera accessory as shown in the figure below.



This function controls whether to enable the macro camera's image preview.(This option only appears when the macro camera hardware is installed)

On: Enables camera preview, but photos are captured in 4K resolution.

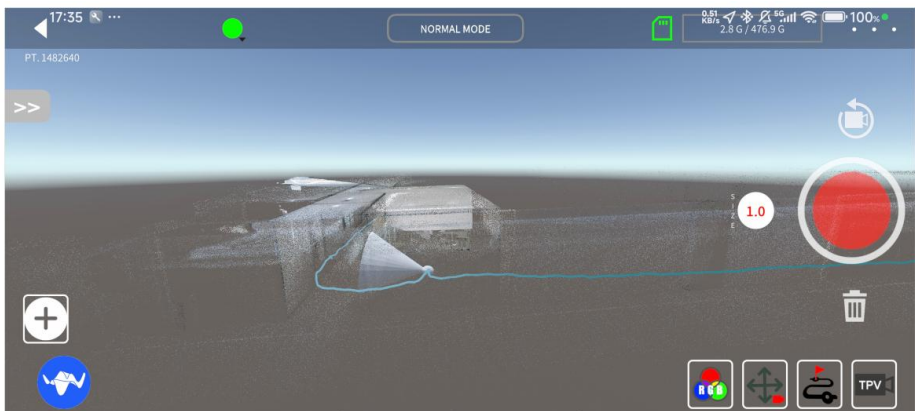
Off: Disables video preview; photos are captured in 8K resolution and

support photo preview.

⚠ When using the detailed camera, the focus time is over 0.5 seconds—hold the device steady before taking photos.

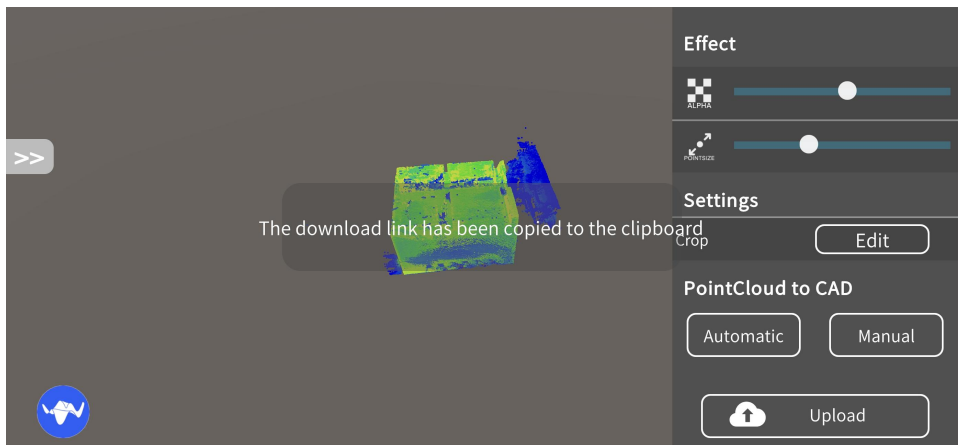


During data collection, pressing the camera button on the macro camera will display the position and direction of the handheld device when taking photos on the point cloud preview page of the APP.



4.7 Upload and direct export of LAS point cloud

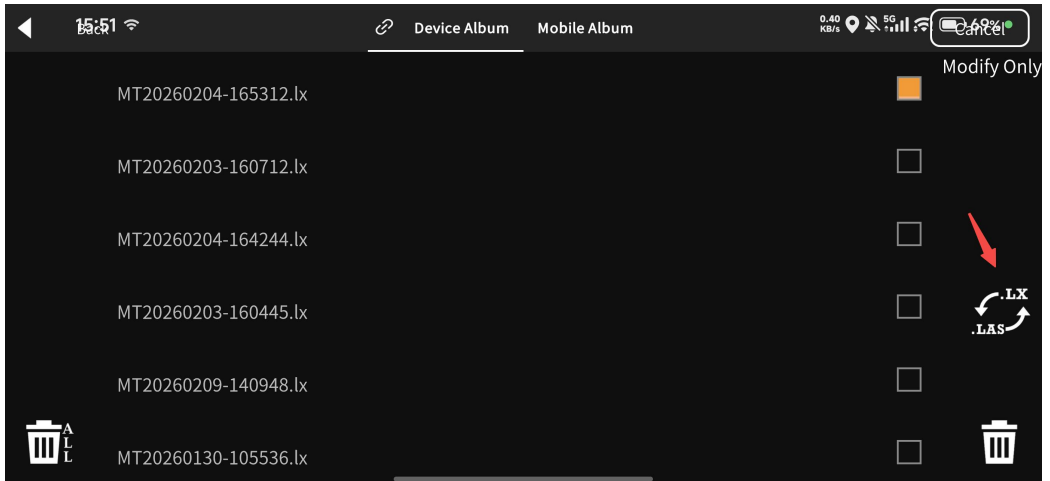
The LAS point cloud upload feature is located in the point cloud file section of a local or device project within the APP's home screen Album. Navigate to the settings page of the point cloud file and tap "Upload" to send the file to the server for conversion to LAS format. Once processed, the app will copy a download link for the converted LAS point cloud to your device's clipboard. Simply open the link to download the file directly.



⚠ The LAS upload feature is only available after logging into your account. The converted point cloud consists of decimated raw data with no additional optimization or processing, and therefore differs significantly in quality from the results obtained through post-processing in the desktop client.

The direct LAS export feature is located in the point cloud file section of a device project within the APP's home screen album. This feature requires the app to be connected to the device's Wi-Fi. Select the .lx file you wish to convert, then tap the format conversion icon indicated by the arrow in the image below.

The LAS point cloud will be generated immediately and saved in the project folder.



⚠ The directly exported LAS point cloud consists of the device's raw scan data without any optimization or post-processing. This conversion feature is suitable for point cloud data from scans covering an area of up to 500 m².

5. Usage Q&A

Q. How to choose the scanning radar mode?

A. Currently, there are Normal and Degraded modes. Use Degraded mode in environments with high feature similarity or low light; use Normal mode otherwise. The mAPPING algorithms differ in logic between modes.

Q. How far can the device be from the phone?

A. The device and phone communicate via Wi-Fi. Considering potential environmental interference and data volume fluctuations, mount the phone on

the device using a holder and avoid separation.

Q. Can point clouds stored on the phone be exported?

A. If "Store on Mobile Device" is enabled in APP settings, you can view historical project point clouds when disconnected. Click "Upload" in the menu to generate a downloadable web link. (Note: Uploaded data is stored on our servers; ensure it is non-confidential.)

Q. What's the difference between the downsampling ratio in radar settings and the preview point limit in display settings?

A. Both reduce mobile performance load for smoother point cloud previews. The downsampling ratio controls points received by the APP; the preview point limit caps displayed points. They can be set separately or together.

6. Troubleshooting Q&A

Q. Unable to find the device Wi-Fi?

A. Restart the device or try with another phone. Check if the environment has many Wi-Fi signals and retry in a different location.

Q. After connecting to the device Wi-Fi, the APP cannot establish a connection?

A. Check if another phone is already connected to the device.

Q. Blank Screen After Launching APP?

A. Check if the APP has location permission on your phone. Restart the APP after granting location access.

Q. Gradual Lagging or Crashes During Scanning?

A. Reduce the "Preview Point Limit" and increase the "Accuracy Value" in the APP's display settings.

Q. Delayed Device Icon/Point Cloud Updates in Preview?

A. If unrelated to phone performance: Restart the device and retest. If unresolved, contact after-sales support.

Q. "Connection Interrupted" Prompt During Scanning?

A. Please turn off your mobile network and ensure that Wifi remains connected all the time. If the automatic reconnection fails in the APP interface, please change your phone for testing and report this phone model to the after-sales service. (For iPhone, please keep the APP always on the front screen and the screen always lit up.)

Q. After the scanning begins or during the scanning process, there is point cloud drift?

A. Make sure the device remains stationary until the initialization pop-up window disappears after clicking the start scanning button. Please restart the device and try again. If the problem persists, please contact the after-sales service.

Q. Yellow Status Indicator in APP?

A. Tap the status indicator to identify the faulty module. If overheating is detected, allow the device to cool before reuse. For other issues, restart the device and test with a different phone. If unresolved, contact after-sales support.

Q. Why isn't the device's storage recognized by the computer when

connected via USB cable?

A. First, ensure the device is powered on. Next, use the original cable provided by the manufacturer or a high-speed data cable that supports USB 3.0. Finally, confirm that both ends of the cable are securely connected.

Q. What charger should I use to charge the device?

A. We recommend using a charger with an output power between 35W and 67W. A steadily blinking blue LED indicates normal charging, and four solid blue LEDs mean the battery is fully charged.

7. App New Version Update Contents

1. Added a warning during firmware flashing: "Do not disconnect power or interact with the device."
2. Implemented proactive OTA firmware update notifications.
3. Added support for NTRIP server domain names (in addition to IP addresses).
4. Improved user interaction when the SD card is full.
5. Enhanced log sharing to allow optional local sharing or uploading to the server.
6. Fixed a series of bugs.